

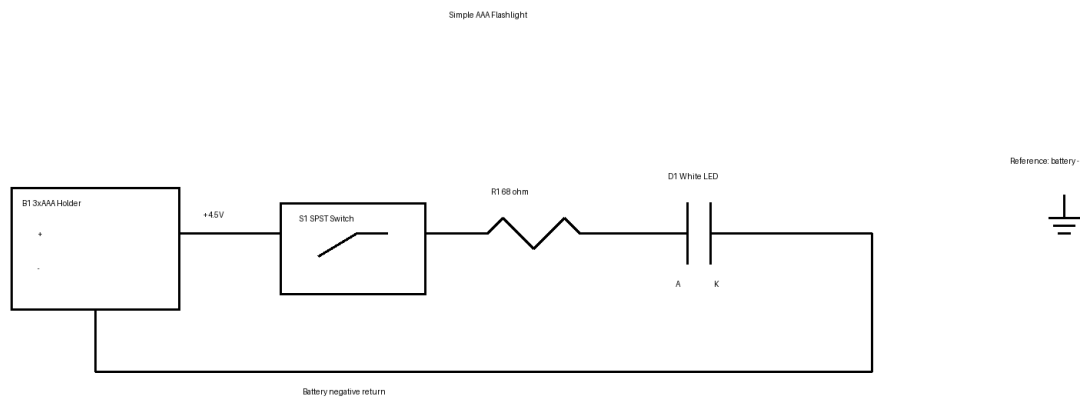
Simple AAA Battery Flashlight

Goal: a simple manual flashlight using three AAA batteries, one switch, one resistor, and one white LED.

Assumptions

- Flashlight uses three AAA alkaline cells in series for a nominal 4.5 V source.
- The flashlight is a simple manual on/off design with no microcontroller.
- LED chosen as a standard white 5 mm LED around 20 mA nominal current.
- A single series resistor is used to set LED current conservatively for typical battery voltage.
- Mechanical enclosure and reflector details are out of scope; this package focuses on the electrical design.

Schematic



Bill of Materials

RefDes	Description	Qty	Unit \$	Ext. \$	Supplier / Link
B1	3x AAA battery holder, series output	1	1.50	1.50	Generic https://www.amazon.com/s?k=3x+aaa+battery+holder
BT1	AAA alkaline battery	3	0.75	2.25	Generic https://www.amazon.com/s?k=aaa+alkaline+batteries
S1	SPST on/off switch	1	0.60	0.60	Generic https://www.amazon.com/s?k=spst+mini+switch
D1	5 mm white LED, Vf ~3.0-3.2 V @ 20 mA	1	0.20	0.20	Generic https://www.amazon.com/s?k=5mm+white+led

R1	68 ohm resistor, 1/4 W	1	0.05	0.05	Generic https://www.digikey.com/en/products/filter/through-hole-resistors/53
J1	Hookup wire / spring contacts / assembly hardware	1	0.75	0.75	Generic https://www.amazon.com/s?k=hookup+wire
	Total estimated cost			5.35	

No firmware is required because this is a purely manual circuit.